Assignment 1

```
Name : Ravish Ranjan
Course : MCA
Semester : 2nd semester
```

TASK 1: Check Local network state

- Step 1: Open Command Prompt/Terminal
- Step 2: Check Your IP Configuration Ans

1. Active Interface

```
ip -o -all addr show | awk '{print $2}'
```

```
ravish@ravishPC:~> ip -o -all addr show | awk '{print $2}'
lo
lo
wlp1s0
wlp1s0
docker0
br-df90e2be7ea0
br-df90e2be7ea0
veth69f7cdc
ravish@ravishPC:~>
```

2. IPv4\IPV6

```
echo -e "\e[32;1mIPV4\e[0m : \n$(ifconfig | grep "inet ")"
```

```
ravish@ravishPC:~> echo -e "\e[32;1mIPV4\e[0m : \n$(ifconfig | grep "inet ")"
IPV4 :
        inet 172.18.0.1  netmask 255.255.0.0  broadcast 172.18.255.255
        inet 172.17.0.1  netmask 255.255.0.0  broadcast 172.17.255.255
        inet 127.0.0.1  netmask 255.0.0.0
        inet 10.85.187.240  netmask 255.255.240.0  broadcast 10.85.191.255
ravish@ravishPC:~>
```

3. Default gateway

```
echo -e "\e[32;1mgateway\e[0m : $(ip route show | grep "default via")"
```

```
ravish@ravishPC:~> echo -e "\e[32;1mgateway\e[0m : $(ip route show | grep "default via")"
gateway : default via 10.85.176.1 dev wlp1s0 proto dhcp src 10.85.187.240 metric 600
ravish@ravishPC:~>
```

4. DNS server

```
cat /etc/resolv.conf
```

```
ravish@ravishPC:~> cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 127.0.0.53
options edns0 trust-ad
search .
ravish@ravishPC:~>
```

5. Why does the output show so many interfaces?

```
I have many interfaces because of many applications which are running on my
system
```

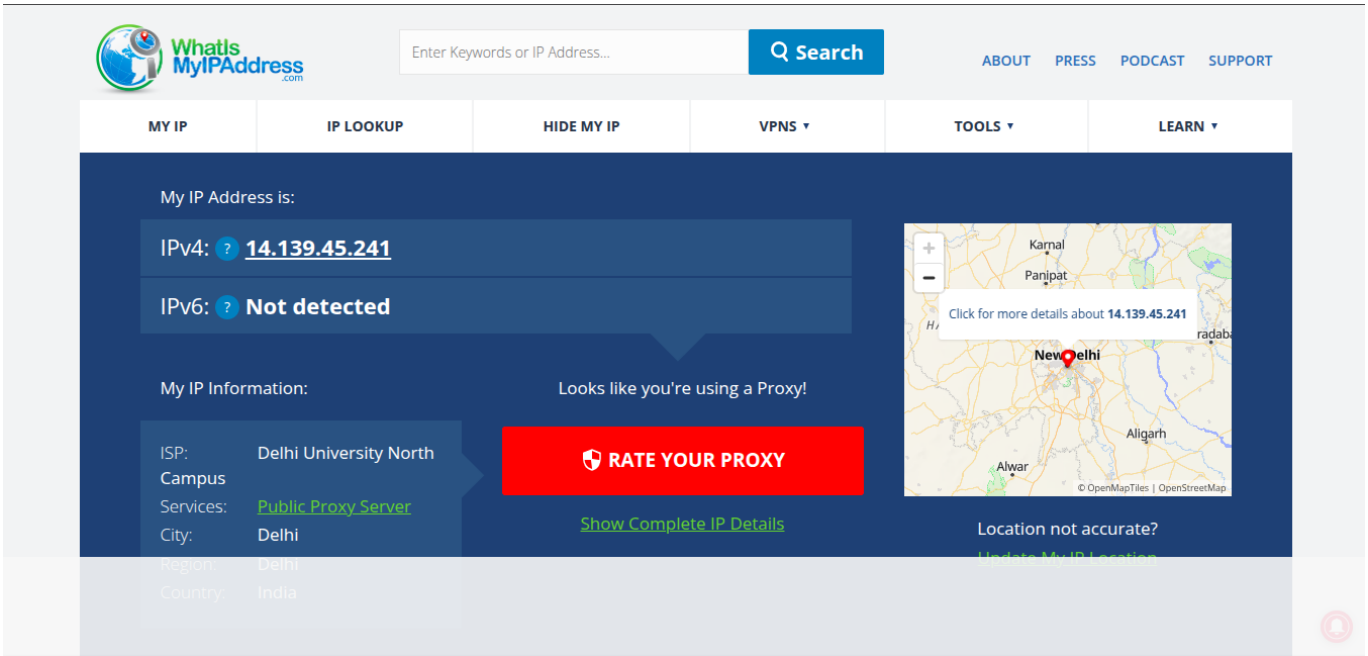
---

6. Determine whether the IP address of your machine is a private IP address or a public IP address?

```
Since my gateway (gateway : 10.85.176.1) is in range 10.0.0.0 to
10.255.255.255  hence it is a private ip address
```

---

7. Identify the public IP address seen from the Internet from any website of your choice.



8. 2-3 lines interpretation of what an attacker could infer.

```
With all this information an attacker could
1. Put me in targets of its attacks
2. Create or find weapons (mechanisms) to attach me
3. Find open ports to deliver the attack
4. Interact and take control of my system though weak links
5. Can detect if i am in a public or private network
6. Find my general location
7. Poison my DNS
8. Ping me repeatedly for DDos attack with found open ports
```

TASK 2: Test Network Connectivity (10 minutes)

Step 1: Run repeated (at least 10) probes to:

1. Gateway (Use the Default Gateway IP you found in Task 1).

```
ravish@ravishPC:~> ping -c 10 10.85.176.1

PING 10.85.176.1 (10.85.176.1) 56(84) bytes of data.
64 bytes from 10.85.176.1: icmp_seq=1 ttl=64 time=2.61 ms
64 bytes from 10.85.176.1: icmp_seq=2 ttl=64 time=38.5 ms
64 bytes from 10.85.176.1: icmp_seq=3 ttl=64 time=2.79 ms
64 bytes from 10.85.176.1: icmp_seq=4 ttl=64 time=77.2 ms
64 bytes from 10.85.176.1: icmp_seq=5 ttl=64 time=3.06 ms
64 bytes from 10.85.176.1: icmp_seq=6 ttl=64 time=3.07 ms
64 bytes from 10.85.176.1: icmp_seq=7 ttl=64 time=4.70 ms
64 bytes from 10.85.176.1: icmp_seq=8 ttl=64 time=1.91 ms
64 bytes from 10.85.176.1: icmp_seq=9 ttl=64 time=3.01 ms
64 bytes from 10.85.176.1: icmp_seq=10 ttl=64 time=2.92 ms

--- 10.85.176.1 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9013ms
rtt min/avg/max/mdev = 1.907/13.974/77.168/23.585 ms
ravish@ravishPC:~>
```

2. 8.8.8.8

```
ravish@ravishPC:~> ping -c 10 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=6.72 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=116 time=5.14 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=116 time=6.94 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=116 time=6.41 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=116 time=6.27 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=116 time=5.77 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=116 time=6.42 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=116 time=5.47 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=116 time=5.31 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=116 time=5.48 ms
--- 8.8.8.8 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9014ms
rtt min/avg/max/mdev = 5.144/5.992/6.941/0.603 ms
ravish@ravishPC:~>
```

3. Any domain of your choice

```
ravish@ravishPC:~> ping -c 10 ravishdev.org
PING ravishdev.org (185.199.108.153) 56(84) bytes of data.
64 bytes from cdn-185-199-108-153.github.com (185.199.108.153): icmp_seq=1 ttl=58 time=6.53 ms
64 bytes from cdn-185-199-108-153.github.com (185.199.108.153): icmp_seq=2 ttl=58 time=6.38 ms
64 bytes from cdn-185-199-108-153.github.com (185.199.108.153): icmp_seq=3 ttl=58 time=7.06 ms
64 bytes from cdn-185-199-108-153.github.com (185.199.108.153): icmp_seq=4 ttl=58 time=8.11 ms
64 bytes from cdn-185-199-108-153.github.com (185.199.108.153): icmp_seq=5 ttl=58 time=6.44 ms
64 bytes from cdn-185-199-108-153.github.com (185.199.108.153): icmp_seq=6 ttl=58 time=14.7 ms
64 bytes from cdn-185-199-108-153.github.com (185.199.108.153): icmp_seq=7 ttl=58 time=6.00 ms
64 bytes from cdn-185-199-108-153.github.com (185.199.108.153): icmp_seq=8 ttl=58 time=6.96 ms
64 bytes from cdn-185-199-108-153.github.com (185.199.108.153): icmp_seq=9 ttl=58 time=11.2 ms
64 bytes from cdn-185-199-108-153.github.com (185.199.108.153): icmp_seq=10 ttl=58 time=11.8 ms

--- ravishdev.org ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9008ms
rtt min/avg/max/mdev = 5.999/8.517/14.668/2.826 ms
ravish@ravishPC:~>
```

to	packet loss	min RTT	avg RTT	max RTT	DNS name resolution affected the probe	explanation for any unusual observation
gateway	0%	1.907 ms	13.974 ms	77.168 ms	no	sudden latency spike due to high traffic
8.8.8.8	0%	5.144 ms	5.992 ms	6.941 ms	no	no unusual observation
ravishdev.org	0%	5.999 ms	8.517 ms	14.668 ms	yes	higher overhead due to dns resolution and low server resources

Step 2: Ping the destination of your choice with 10 echo Request messages, each of which has a Data field of 1000 bytes. Take a screenshot of the output.

```
ravish@ravishPC:~> ping -c 10 -s 1000 ravishdev.org
PING ravishdev.org (185.199.109.153) 1000(1028) bytes of data.
1008 bytes from cdn-185-199-109-153.github.com (185.199.109.153): icmp_seq=1 ttl=58 time=6.62 ms
1008 bytes from cdn-185-199-109-153.github.com (185.199.109.153): icmp_seq=2 ttl=58 time=6.48 ms
1008 bytes from cdn-185-199-109-153.github.com (185.199.109.153): icmp_seq=3 ttl=58 time=7.14 ms
1008 bytes from cdn-185-199-109-153.github.com (185.199.109.153): icmp_seq=4 ttl=58 time=6.95 ms
1008 bytes from cdn-185-199-109-153.github.com (185.199.109.153): icmp_seq=5 ttl=58 time=6.36 ms
1008 bytes from cdn-185-199-109-153.github.com (185.199.109.153): icmp_seq=6 ttl=58 time=7.28 ms
1008 bytes from cdn-185-199-109-153.github.com (185.199.109.153): icmp_seq=7 ttl=58 time=6.92 ms
1008 bytes from cdn-185-199-109-153.github.com (185.199.109.153): icmp_seq=8 ttl=58 time=9.75 ms
1008 bytes from cdn-185-199-109-153.github.com (185.199.109.153): icmp_seq=9 ttl=58 time=7.57 ms
1008 bytes from cdn-185-199-109-153.github.com (185.199.109.153): icmp_seq=10 ttl=58 time=5.78 ms

--- ravishdev.org ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9012ms
rtt min/avg/max/mdev = 5.775/7.084/9.750/1.012 ms
ravish@ravishPC:~>
```

TASK 3 Comparative Route Analysis

Step 1: Run traceroute/ tracert/ pathping (optional route analysis tool) to three destinations (domain)

1. google.com

```
ravish@ravishPC:~> traceroute google.com
traceroute to google.com (142.251.223.142), 30 hops max, 60 byte packets
 1  _gateway (192.168.1.1)  4.026 ms  3.885 ms  3.822 ms
 2  10.240.15.12 (10.240.15.12)  5.762 ms  5.722 ms  5.680 ms
 3  * * *
 4  152.52.174.61 (152.52.174.61)  7.400 ms aes-static-021.133.22.125.airtel.in (125.22.133.21)  5.903 ms 152.52.174.61 (152.52.174.61
)  7.319 ms
 5  142.250.168.22 (142.250.168.22)  7.278 ms 116.119.121.60 (116.119.121.60)  6.744 ms  21.218 ms
 6  * * *
 7  * * 142.251.54.64 (142.251.54.64)  7.492 ms
 8  142.251.76.198 (142.251.76.198)  7.832 ms 142.251.54.100 (142.251.54.100)  7.042 ms 142.251.54.85 (142.251.54.85)  6.323 ms
 9  tzdela-be-in-f14.1e100.net (142.251.223.142)  6.471 ms 142.251.54.83 (142.251.54.83)  8.614 ms  8.578 ms
ravish@ravishPC:~> ^C
```

2. youtube.com

```
ravish@ravishPC:~> traceroute youtube.com
traceroute to youtube.com (142.250.207.142), 30 hops max, 60 byte packets
 1  _gateway (192.168.1.1)  5.652 ms  5.594 ms  5.559 ms
 2  10.240.15.12 (10.240.15.12)  13.562 ms  13.529 ms  13.498 ms
 3  * * *
 4  aes-static-021.133.22.125.airtel.in (125.22.133.21)  9.305 ms 152.52.174.61 (152.52.174.61)  9.506 ms aes-static-021.133.22.125.ai
rtel.in (125.22.133.21)  9.474 ms
 5  116.119.121.60 (116.119.121.60)  9.439 ms 142.250.168.22 (142.250.168.22)  10.558 ms  10.522 ms
 6  * * *
 7  216.239.49.36 (216.239.49.36)  8.548 ms 209.85.252.44 (209.85.252.44)  9.836 ms 142.251.52.198 (142.251.52.198)  8.706 ms
 8  192.178.83.206 (192.178.83.206)  8.666 ms 172.253.50.156 (172.253.50.156)  9.202 ms 142.251.52.218 (142.251.52.218)  9.953 ms
 9  192.178.83.220 (192.178.83.220)  44.984 ms 192.178.46.225 (192.178.46.225)  10.270 ms 142.251.255.56 (142.251.255.56)  8.491 ms
10  192.178.252.109 (192.178.252.109)  9.503 ms 142.251.197.114 (142.251.197.114)  34.893 ms 192.178.46.225 (192.178.46.225)  17.474 m
s
11  192.178.254.238 (192.178.254.238)  28.416 ms 142.251.198.54 (142.251.198.54)  31.074 ms 192.178.254.212 (192.178.254.212)  33.969
ms
12  142.250.226.135 (142.250.226.135)  31.895 ms 192.178.253.211 (192.178.253.211)  29.863 ms 142.250.212.86 (142.250.212.86)  30.709
ms
13  192.178.110.249 (192.178.110.249)  28.390 ms 142.250.214.101 (142.250.214.101)  30.380 ms 142.250.214.99 (142.250.214.99)  32.039
ms
14  pnbomb-bb-in-f14.1e100.net (142.250.207.142)  31.563 ms 142.250.214.101 (142.250.214.101)  29.970 ms pnbomb-bb-in-f14.1e100.net (1
42.250.207.142)  31.836 ms
ravish@ravishPC:~>
```

3. ravishdev.org

```
ravish@ravishPC:~> traceroute ravishdev.org
traceroute to ravishdev.org (185.199.109.153), 30 hops max, 60 byte packets
 1  _gateway (192.168.1.1)  1.948 ms  5.388 ms  5.446 ms
 2  10.240.15.12 (10.240.15.12)  11.972 ms  11.931 ms  11.893 ms
 3  172.31.0.194 (172.31.0.194)  13.389 ms * *
 4  152.52.174.61 (152.52.174.61)  11.644 ms  11.612 ms  11.570 ms
 5  182.79.235.253 (182.79.235.253)  19.951 ms  11.485 ms  11.453 ms
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
ravish@ravishPC:~>
```

to	Total no. of hops	first hop	second hop	final destination
google.com	9	192.168.1.1	10.240.15.12	142.251.223.142
youtube.com	14	192.168.1.1	10.240.15.12	142.250.207.142
ravishdev.org	30	192.168.1.1	10.240.15.12	server refused to repond

Questions

1. Why might some hops show *** request timed out even when the destination is reachable?

```
- Path still works; only visibility is blocked
- Rate limiting on intermediate devices
- Server refused to repond
```

2. What can an attacker infer from traceroute results about a target network?

```
- Network path and intermediate providers
- Approximate geographic routing
- Potential choke points or weak segments
- Internal IP ranges
```

3. Why is the first hop usually your local router or gateway?

```
Since at first hop ttl of package is 1 it it expires at only first hop
which is the default gateway (router)
```

4. Why may the second or third hop often belong to the ISP?

```
After router the next hop is generally to the ISP (if over wifi) via
which all trafic is routed
```

5. How could traceroute help in identifying whether a service is hosted locally, in another region, or behind a cloud provider?

```
We can you the analyzed latency and cloud providers name to figure out
the region and providers
generally:
    - lo latency/fewer hops -> local hosting
    - Sudden latency jump -> different region/country
    - Well-known ASN names -> AWS/GCP/Azure
    - Anycast/CDN patterns -> multiple similar endpoints
```

6. Why is traceroute useful to both defenders and attackers?

```
For defenders:
    - diagnose latency and routing issues
    - identifing the failing links

For attackers:
    - mapping network topology
    - recconaissence of target
    - planning delivery path of attack
```

TASK 4: DNS footprint analysis

Select two public domains for analysis:

- One large service/platform domain, such as github.com or google.com

`amazon.com`

- One institutional, academic, or organizational domain.

`linux.org`

Step 1: Perform a basic DNS Lookup to verify that the domain resolves successfully.

1. amazon.com

```
Server:     127.0.0.53
Address:    127.0.0.53#53

Non-authoritative answer:
Name:   amazon.com
Address: 98.87.170.74
Name:   amazon.com
Address: 98.82.161.185
Name:   amazon.com
Address: 98.87.170.71
```

2. linux.org

```
Server:     127.0.0.53
Address:    127.0.0.53#53

Non-authoritative answer:
Name:   linux.org
Address: 104.26.15.72
Name:   linux.org
Address: 172.67.73.26
Name:   linux.org
Address: 104.26.14.72
Name:   linux.org
Address: 2606:4700:839a:b04b:68a6:83f:2bb6:48cf
```

Step 2: Run the same query multiple times. Observe whether the result suggests single-hostresolution, redundancy, or load balancing.

to	hosting type	reason
amazon.com	load balancing	different order of same multiple ips of each query of nslookup
linux.org	load balancing	different order of same multiple ips of each query of nslookup

Step 3: For each of the two domains, collect the following DNS record types:

to	amazon.com	linux.org
A	Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: Name: amazon.com Address: 98.87.170.71 Name: amazon.com Address: 98.82.161.185 Name: amazon.com Address: 98.87.170.74	ravish@ravishPC:~> nslookup -type=A linux.org Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: Name: linux.org Address: 104.26.15.72 Name: linux.org Address: 104.26.14.72 Name: linux.org Address: 172.67.73.26
AAAA	Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: *** Can't find amazon.com: No answer	ravish@ravishPC:~> nslookup -type=AAAA linux.org Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: Name: linux.org Address: 2606:4700:839a:b04b:68ef:83f:2bb6:48cf
NS	Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: amazon.com nameserver = ns1.amzndns.co.uk. amazon.com nameserver = ns1.amzndns.com. amazon.com nameserver = ns1.amzndns.net. amazon.com nameserver = ns1.amzndns.org. amazon.com nameserver = ns2.amzndns.co.uk. amazon.com nameserver = ns2.amzndns.com. amazon.com nameserver = ns2.amzndns.net. amazon.com nameserver = ns2.amzndns.org. Authoritative answers can be found from:	ravish@ravishPC:~> nslookup -type=NS linux.org Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: linux.org nameserver = lia.ns.cloudflare.com. linux.org nameserver = mark.ns.cloudflare.com. Authoritative answers can be found from:
MX	Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: amazon.com mail exchanger = 5 amazon-smtp.amazon.com. Authoritative answers can be found from: amazon.com nameserver = ns1.amzndns.co.uk. amazon.com nameserver = ns1.amzndns.com. amazon.com nameserver = ns1.amzndns.net. amazon.com nameserver = ns1.amzndns.org. amazon.com nameserver = ns2.amzndns.co.uk. amazon.com nameserver = ns2.amzndns.com. amazon.com nameserver = ns2.amzndns.net. amazon.com nameserver = ns2.amzndns.org.	ravish@ravishPC:~> nslookup -type=MX linux.org Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: linux.org mail exchanger = 10 mx1.improvmx.com. linux.org mail exchanger = 20 mx2.improvmx.com. Authoritative answers can be found from:
TXT	ravish@ravishPC:~> nslookup -type=TXT amazon.com ;; communications error to 127.0.0.53#53: timed out ;; communications error to 127.0.0.53#53: timed out ;; communications error to 127.0.0.53#53: timed out ;; no servers could be reached	ravish@ravishPC:~> nslookup -type=TXT linux.org Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: linux.org text = "ahrefs-site-verification_631032cd30b4e19d3273793b1c65f11c66f6a5426ba38536b9d58ef0dc91054d" Authoritative answers can be found from:
reverse lookup	ravish@ravishPC:~> nslookup 98.87.170.71 71.170.87.98.in-addr.arpa name = ec2-98-87-170-71.compute-1.amazonaws.com. Authoritative answers can be found from: 170.87.98.in-addr.arpa nameserver = ns1-24-us-east-1.ec2-rdns.amazonaws.com. 170.87.98.in-addr.arpa nameserver = ns2-24-us-east-1.ec2-rdns.amazonaws.com. 170.87.98.in-addr.arpa nameserver = ns3-24-us-east-1.ec2-rdns.amazonaws.com. 170.87.98.in-addr.arpa nameserver = ns4-24-us-east-1.ec2-rdns.amazonaws.com.	ravish@ravishPC:~> nslookup 104.26.15.72 ** server can't find 72.15.26.104.in-addr.arpa: NXDOMAIN ravish@ravishPC:~>

Step 4: for each of the domains, create a table in the following format.

amazon.com

Record Type	Observed Output	Technical Meaning	Reconnaissance Value
A	```Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: Name: amazon.com Address: 98.87.170.71 Name: amazon.com Address: 98.82.161.185 Name: amazon.com Address: 98.87.170.74	load balanced or redundant servers	Reveals server IP Helps identify hosting provider Enables port scanning/geo-location
AAAA	```Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: *** Can't find amazon.com: No answer	no IPv6 assigned	no record
NS	```Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: amazon.com nameserver = ns1.amzndns.co.uk. amazon.com nameserver = ns1.amzndns.com. amazon.com nameserver = ns1.amzndns.net. amazon.com nameserver = ns1.amzndns.org. amazon.com nameserver = ns2.amzndns.co.uk. amazon.com nameserver = ns2.amzndns.com. amazon.com nameserver = ns2.amzndns.net. amazon.com nameserver = ns2.amzndns.org. Authoritative answers can be found from:	have redundant and reliable nameservers	Shows DNS provider Helps identify infrastructure control points
MX	```Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: amazon.com mail exchanger = 5 amazon-smtp.amazon.com. Authoritative answers can be found from: amazon.com nameserver = ns1.amzndns.co.uk. amazon.com nameserver = ns1.amzndns.com. amazon.com nameserver = ns1.amzndns.net. amazon.com nameserver = ns1.amzndns.org. amazon.com nameserver = ns2.amzndns.co.uk. amazon.com nameserver = ns2.amzndns.com. amazon.com nameserver = ns2.amzndns.net. amazon.com nameserver = ns2.amzndns.org.	one mail server with priority 5	Shows email infrastructure Useful for phishing targeting / email attack surface
TXT	```ravish@ravishPC:~> nslookup -type=TXT amazon.com ;; communications error to 127.0.0.53#53: timed out ;; communications error to 127.0.0.53#53: timed out ;; communications error to 127.0.0.53#53: timed out ;; no servers could be reached	no field set for arbitary data authorization and domain verification	May expose internal services or integrations Can hint at cloud providers or tools used
Reverse Lookup	```ravish@ravishPC:~> nslookup 98.87.170.71 71.170.87.98.in-addr.arpa name = ec2-98-87-170-71.compute-1.amazonaws.com. Authoritative answers can be found from: 170.87.98.in-addr.arpa nameserver = ns1-24-us-east-1.ec2-rdns.amazonaws.com. 170.87.98.in-addr.arpa nameserver = ns2-24-us-east-1.ec2-rdns.amazonaws.com. 170.87.98.in-addr.arpa nameserver = ns3-24-us-east-1.ec2-rdns.amazonaws.com. 170.87.98.in-addr.arpa nameserver = ns4-24-us-east-1.ec2-rdns.amazonaws.com.	ip is mapped to the domain name usually for loggin and verification	Can expose server roles or environment

linux.org

Record Type	Observed Output	Technical Meaning	Reconnaissance Value

A	`ravish@ravishPC:~> nslookup -type=A linux.org`
	`Server: 127.0.0.53`
	`Address: 127.0.0.53#53`
	
	`Non-authoritative answer:`
	`Name: linux.org`
	`Address: 104.26.15.72`
	`Name: linux.org`
	`Address: 104.26.14.72`
	`Name: linux.org`
	`Address: 172.67.73.26`
		load balanced or redundant servers	Reveals server IP
			Helps identify hosting provider
			Enables port scanning/geo-location

AAAA	`ravish@ravishPC:~> nslookup -type=AAAA linux.org`
	`Server: 127.0.0.53`
	`Address: 127.0.0.53#53`
	
	`Non-authoritative answer:`
	`Name: linux.org`
	`Address: 2606:4700:839a:b04b:68ef:83f:2bb6:48cf`
		one IPv6 assigned	Reveals server IP
			Helps identify hosting provider
			Enables port scanning/geo-location

NS	`ravish@ravishPC:~> nslookup -type=NS linux.org`
	`Server: 127.0.0.53`
	`Address: 127.0.0.53#53`
	
	`Non-authoritative answer:`
	`linux.org nameserver = lia.ns.cloudflare.com.`
	`linux.org nameserver = mark.ns.cloudflare.com.`
	
	`Authoritative answers can be found from:`
		have redundant and reliable nameservers	Shows DNS provider
			Helps identify infrastructure control points

MX	`ravish@ravishPC:~> nslookup -type=MX linux.org`
	`Server: 127.0.0.53`
	`Address: 127.0.0.53#53`
	
	`Non-authoritative answer:`
	`linux.org mail exchanger = 10 mx1.improvmx.com.`
	`linux.org mail exchanger = 20 mx2.improvmx.com.`
	
	`Authoritative answers can be found from:`
		2 mail servers with priority 10 and 20	Shows email infrastructure
			Useful for phishing targeting / email attack surface

TXT	`ravish@ravishPC:~> nslookup -type=TXT linux.org`
	`Server: 127.0.0.53`
	`Address: 127.0.0.53#53`
	
	`Non-authoritative answer:`
	`linux.org text = "ahrefs-site-verification_631032cd30b4e19d3273793b1c65f11c66f6a5426ba38536b9d58ef0dc91054d"`
	
	`Authoritative answers can be found from:`
		one field set for arbitary data authorization and domain verification	May expose internal services or integrations
			Can hint at cloud providers or tools used

Reverse Lookup	`ravish@ravishPC:~> nslookup 104.26.15.72`
	`** server can't find 72.15.26.104.in-addr.arpa: NXDOMAIN`
	`ravish@ravishPC:~>`
		reverse lookup not set	no record

Step 5: Identify which one exposes more useful reconnaissance information and why.

`linux.org` exposes more reconnaissance information than `amazon.com`.

1. linux.org provides a visible TXT record (site verification), which reveals information about domain verification and possible integrations, whereas amazon.com does not return useful TXT data in this

analysis.

2. The IP addresses of linux.org are relatively fewer and associated with Cloudflare, making it easier to infer the hosting/CDN provider, while amazon.com uses highly distributed AWS infrastructure with multiple IPs, making direct mapping harder.

3. The NS and MX records of linux.org clearly indicate third-party services (eg. Cloudflare for DNS and ImprovMX for email), which reveals external dependencies, whereas amazon.com uses its own infrastructure (Amazon DNS and mail), reducing exposed information.

4. amazon.com uses large-scale load balancing and infrastructure abstraction, which hides internal architecture details, while linux.org exposes more direct service relationships.

Therefore, linux.org provides more accessible and interpretable reconnaissance data compared to amazon.com.